Topological Persistent Homology of Diffusion-Weighted MRI: A Model-Free Non-Gaussianity Biomarker Linked to Lévy-Stable Displacement Statistics

Chisondi S. Warioba*[1]

[1]Division of Pain Medicine, Department of Anesthesiology, Perioperative and Pain Medicine, Stanford University School of Medicine, Stanford, California 94305, USA. ORCID: 0000-0002-4266-2673.

Submitted to *Magnetic Resonance in Medicine*
Version of May 21, 2026

Abstract

Purpose: To introduce a topological non-Gaussianity biomarker for diffusion-weighted magnetic resonance imaging (DW-MRI) derived from the sublevel-set persistent homology of multi-shell, multi-direction signals, and to relate it to the Lévy-stable parametrisation of the molecular displacement distribution.

Theory and Methods: For a voxel population in which the underlying molecular displacement distribution is symmetric α-stable with index $\alpha \in (0, 2)$, the pooled centred log-attenuation values $-\ln(S/S_0)$ across all voxels in an anatomically homogeneous region of interest (ROI) and all gradient directions form, after cumulative summation, a discrete-time approximation of an α-stable Lévy bridge. Sublevel-set persistent homology of this bridge has lifetimes whose tail is regularly varying with exponent α (see Theorem 1); the Hill estimator on the top lifetimes is consistent for α. We benchmark this ROI-level $\widehat{\alpha}_{\mathrm{persistence}}$ against diffusion-kurtosis K (DKI) and the stretched-exponent $\alpha_{\mathrm{se}}/2$ on a forward model containing free Gaussian, restricted Gaussian, α-stable, and mixed compartments.

Results: On simulated alpha-stable sample paths the persistence-tail estimator recovers the true stability index with mean bias 0.03 and root-mean-square error 0.10 at $n = 1000$ steps. Applied to $N = 30$ Human Connectome Project subjects, the ROI-level $\widehat{\alpha}_{\mathrm{persistence}}$ orders tissue classes robustly (mean across subjects: CC$= 2.19$, WM$= 1.95$, GM$= 1.03$, CSF$= 0.85$; one-way ANOVA $F \gg 100$; pairwise paired permutation $p \leq 10^{-4}$ at 10,000 permutations). On $N = 44$ HCP-YA Retest subjects, scan-rescan reliability shows a regime dichotomy predicted by the parent theorem: in the heavy-tailed regime (alpha well below 2), GM intraclass correlation ICC$(3, 1) = 0.55$ and CSF ICC$= 0.63$, broadly comparable to DKI mean kurtosis on the same data (CSF MK ICC$= 0.86$, GM MK$= 0.61$); in the Gaussian-boundary regime (alpha approaching 2, WM and CC), the Hill estimator is degenerate by construction

*Corresponding author: `cwarioba@stanford.edu`

and per-subject ICC drops to near zero, consistent with the exponential-to-power-law transition at the Gaussian boundary. $\widehat{\alpha}_{\text{persistence}}$ therefore provides a per-subject biomarker of non-Gaussian molecular displacement in the heavy-tail regime and a model-free tissue-class characterisation everywhere.

Conclusion: The persistence-tail framework supplies a geometric, model-free non-Gaussianity descriptor of DW-MRI signals whose theoretical foundation is the limit theorem for sample-path topology of α-stable processes. SLURM-ready code that produces voxelwise maps of $\widehat{\alpha}_{\text{persistence}}$, $\widehat{K}$, and $\widehat{\alpha}_{\text{se}}$ from Human Connectome Project and Massachusetts General Hospital Adult Diffusion datasets is released with the manuscript.

Keywords: diffusion MRI; non-Gaussian diffusion; topological data analysis; persistent homology; Lévy processes; microstructure imaging.

1 Introduction

Diffusion-weighted MRI probes tissue microstructure through the attenuation of the spin-echo signal as a function of the diffusion weighting b:

$$S(b, \hat{g}) = S_0 \, \mathbb{E}[\exp(i \, q \, \hat{g} \cdot \mathbf{r})] \,, \qquad q = \sqrt{b/\Delta}, \tag{1}$$

where $\mathbf{r}$ is the molecular displacement during the diffusion time Δ and $\hat{g}$ is the gradient direction. For purely Gaussian displacements the Stejskal–Tanner model $S(b) = S_0 \exp(-bD)$ recovers the apparent diffusion coefficient D [16]. Real tissue, however, is far from Gaussian: cell membranes, axonal restriction, and intra-voxel heterogeneity all introduce heavier-than-Gaussian displacement tails [7, 10, 12].

The two dominant non-Gaussian descriptors in clinical use are diffusion kurtosis imaging [9]

$$\ln S(b) = \ln S_0 - bD + \tfrac{1}{6} \, b^2 D^2 K, \tag{2}$$

and the stretched-exponential model [1]

$$S(b) = S_0 \, \exp\Big[-(bD)^{\alpha_{\text{se}}/2}\Big] \,. \tag{3}$$

DKI captures the fourth moment K of the displacement distribution; the stretched exponent $\alpha_{\text{se}}/2$ provides a single-shape parameter that interpolates between $\alpha_{\text{se}} = 2$ (Gaussian) and $\alpha_{\text{se}} \to 0$ (sharp restriction). Both descriptors require a parametric ansatz and either suppress higher moments (DKI) or constrain the family of admissible tails (stretched exponential).

A complementary route, established by Magin et al. [10], models the displacement distribution as a 3D isotropic symmetric α-stable density with characteristic function $\exp(-(D_\alpha \Delta)^{\alpha/2} |\mathbf{q}|^\alpha)$, recovering

$$S(b) = S_0 \, \exp\Big[-(b \, D_\alpha)^{\alpha/2}\Big] \,, \tag{4}$$

in which the stability index $\alpha \in (0, 2]$ encodes the heaviness of the displacement tail. Equation (4) reduces to Gaussian diffusion at $\alpha = 2$ and reproduces the stretched-exponential form with $\alpha_{\text{se}} = \alpha$. The Lévy-stable model is attractive because it isolates a single, physically meaningful parameter (α) that controls the entire shape of the displacement propagator, but recovering α from a small number of b-shells is a delicate non-linear fitting problem.

Topological framework. In Warioba [21] (hereafter the parent paper) we proved that the sublevel-set persistent homology of an α-stable Lévy sample path has a power-law lifetime tail with exponent equal to α:

$$\frac{1}{N}\,\#\Big\{i:\,\widetilde{\ell}_i > x\Big\} \;\xrightarrow{\mathbb{P}}\; C_\alpha\, x^{-\alpha}, \qquad x \to \infty, \tag{5}$$

with an exponential tail recovered at $\alpha = 2$. Equation (9) gives a topological–phase transition at $\alpha = 2$ and a nonparametric Hill-type estimator,

$$\widehat{\alpha}_{\text{tail}} = \left[\frac{1}{k}\sum_{i=1}^{k}\log\frac{\ell_{(N-i+1)}}{\ell_{(N-k)}}\right]^{-1}, \tag{6}$$

that uses persistence lifetimes rather than raw increments. The estimator is consistent on simulated paths and discriminates classes of anomalous diffusion at trajectory lengths as small as $N = 100$.

Contribution of this paper. The parent paper's discussion mentions DW-MRI as a target application but does not develop the connection. The present manuscript develops the connection in three steps. First, we derive an explicit correspondence between the per-direction attenuation in a single HARDI shell and the sample-path topology of a symmetric α-stable Lévy bridge, justifying the application of (6) to multi-shell DW-MRI signals (Section 2). Second, we introduce a region-pooled persistence-tail estimator and benchmark it against DKI $\widehat{K}$ and stretched $\widehat{\alpha}_{\text{se}}$ on a forward model that contains free Gaussian, restricted Gaussian, α-stable, and mixed compartments (Section 4). Third, we provide SLURM-ready code that produces voxelwise persistence-tail maps from HCP and MGH Adult Diffusion datasets (Section 5).

2 Theory

2.1 Persistence of a one-dimensional path

For a 1D function $f : \{0, 1, \ldots, n\} \to \mathbb{R}$, sublevel-set persistent homology in degree zero pairs each local minimum (a "birth") with the smallest function value at which the basin merges with a deeper one (its "death"). The lifetime of a feature is $\ell = d - b$. The persistence diagram $\mathrm{PD}(f)$ is the multiset of (b, d) pairs. The pairing is determined by the elder rule and can be computed in $O(n \log n)$ time with a union-find data structure [5].

2.2 From DW-MRI signals to Lévy bridges

Let Ω be an anatomically homogeneous region of interest (ROI) containing V voxels, indexed $v = 1, \ldots, V$. For each voxel and each non-zero b-shell $b_k \in \{b_1, \ldots, b_K\}$ with n_k gradient directions, define the per-direction attenuation

$$y_{v,k,i} = -\ln(S_v(b_k, \hat{g}_i)/S_{v,0}), \quad i = 1, \ldots, n_k. \tag{7}$$

Under an isotropic alpha-stable displacement model (Equation 4), the noise-free attenuation has only weak angular structure within Ω; the $y_{v,k,i}$ fluctuate around a ROI-level mean because of (a) Rician measurement noise, (b) finite-ensemble molecular sampling, and (c) microstructural variation across Ω. After centring all Vn_k values per shell to their pooled mean, the row-major concatenation

$$Z_{k,m} = \sum_{m'=1}^{m} \left(y_{v(m'),k,i(m')} - \overline{y_k} \right), \quad m = 1, \ldots, Vn_k, \tag{8}$$

defines a long discrete bridge ($Z_{k,0} = Z_{k,Vn_k} = 0$). When the underlying displacement distribution has a power-law tail of index $\alpha \in (0, 2)$ and finite mean, the generalised central limit theorem implies that $Z_{k,\cdot}$, suitably rescaled, converges in distribution to a symmetric α-stable Lévy bridge of length Vn_k [2]. ROI-level averaging preserves the stability index: averaging V independent α-stable variates yields another α-stable variate with the same α and a reduced scale [13].

The next result, proved in companion work [21], characterises the persistence-lifetime tail of such bridges.

Theorem 1 (Lifetime tail exponent; Warioba 21)**.** *Let L_t, $0 \le t \le T$, be a symmetric α-stable Lévy process with $\alpha \in (0, 2)$, observed on a grid of N points with spacing $\Delta = T/N$. Let $\widetilde{\ell}_i = \ell_i/\Delta^{1/\alpha}$ be the rescaled sublevel-set persistence lifetimes. Then*

$$\frac{1}{N} \# \left\{ i : \widetilde{\ell}_i > x \right\} \xrightarrow{\mathbb{P}} C_\alpha \, x^{-\alpha}, \qquad x \to \infty, \tag{9}$$

for an explicit constant $C_\alpha > 0$. At $\alpha = 2$ the corresponding limit has exponential tails (no power law).

Proof sketch. Each finite persistence pair corresponds to an excursion of the path away from a running minimum. For $\alpha < 2$ the marginal jump size of L_t has a regularly varying tail with exponent α, so excursion heights inherit a tail of the same exponent by the Hewitt–Savage zero-one law and a coupling to the underlying Poisson point process of jumps. A discretisation argument shows that the Skorokhod-grid persistence lifetimes recover this tail with bounded relative error in the grid spacing. A complete proof and a derivation of C_α are in Warioba [21, Appendix A].

Remark 2 (Gaussian-boundary degeneracy)**.** *Theorem 1 excludes $\alpha = 2$ because the lifetime tail of a Brownian bridge is exponential, not power-law. The Hill estimator* (6) *is therefore not consistent at the Gaussian boundary; on data with $\alpha \to 2$ it formally diverges, while finite-sample $\widehat{\alpha}_{\text{tail}}$ saturates near* 2*. This boundary degeneracy is a feature of the theory and is borne out empirically in the in-vivo analysis of Section 5, where ROIs whose ground-truth α approaches* 2 *(white matter, corpus callosum) lose per-subject reproducibility while ROIs with α well inside* $(0, 2)$ *(grey matter, CSF) remain reproducible.*

2.3 Region pooling

Each shell contributes Vn_k centred increments to (8), giving sublevel-set persistence diagrams with order Vn_k finite pairs. Pooling lifetimes across shells before the Hill estimator multiplies the effective sample size by K and suppresses voxelwise noise by $1/\sqrt{V}$ in each increment; this is the construction we use throughout the in-vivo analysis. Voxelwise estimation is recovered in the limit $V \to 1$ but is unreliable at HCP-like SNR (see Section 5).

2.4 Relation to kurtosis and the stretched exponent

For an alpha-stable propagator with index α, the displacement moments of order $q < \alpha$ are finite while moments of order $q \geq \alpha$ diverge. The DKI kurtosis K is therefore not formally defined at $\alpha < 2$; the regression (2) truncated to a finite $b_{\max}$ still produces a finite $\widehat{K}$ that varies smoothly with α. The stretched exponent α_{se} equals α exactly when the underlying propagator is alpha-stable, but mis-specification (mixtures, restriction, kurtosis-only deviation) yields biased $\widehat{\alpha}_{\mathrm{se}}$. The persistence-tail estimator (6) does not require a parametric ansatz on the propagator and is the only of the three quantities that is defined directly in terms of sample-path topology.

3 Methods: simulation pipeline

3.1 Forward models

We implement six DW-MRI forward models in Python on top of the parent paper's α-stable generator and persistence library:

1. **Free Gaussian**: $S(b) = S_0\, e^{-bD}$, $D \in \{0.7, 0.8, 1.0\} \times 10^{-3}$ mm^2/s.
2. **DKI**: $\ln S(b) = \ln S_0 - bD + \frac{1}{6} b^2 D^2 K$, $K \in \{0.5, 1.0, 1.5\}$.
3. **Stretched exponential**: $S(b) = S_0\, e^{-(bD)^{\alpha_{\mathrm{se}}/2}}$ with $\alpha_{\mathrm{se}} \in \{1.0, 1.5, 1.8\}$.
4. **Restricted cylinders**: Gaussian-phase approximation [17], radius $R \in \{1, 2, 5\}$ µm, intra-cylinder diffusivity $D_{\mathrm{intra}} = 1.7 \times 10^{-3}$ mm^2/s, parallel/perpendicular diffusivities derived from saturating mean-squared displacement at the given Δ.
5. **Restricted spheres**: long-Δ limit, isotropic.
6. **Alpha-stable propagator**: per (4) with $\alpha \in \{1.2, 1.5, 1.8, 1.9\}$.

For each model the displacement is sampled at the molecular level (1D or 3D as appropriate) using the parent paper's Chambers–Mallows–Stuck generator, then the noise-free signal is evaluated either through the closed-form attenuation or through the empirical characteristic function $\widehat{S}(b, \hat{g}) = N^{-1} \sum_i \cos(q\, \hat{g} \cdot \mathbf{r}_i)$.

3.2 HARDI acquisition simulation

We simulate Human Connectome Project-style multi-shell acquisitions: $b \in \{0, 1000, 2000, 3000\}$ s/mm^2, 90 Fibonacci-sphere directions per shell, $\Delta = 50$ ms. Rician noise with SNR $= 30$ at $b = 0$ is added through the standard magnitude-image construction $|S + n_R + in_I|$. The output of one voxel simulation is a real array of shape $(n_{\text{shells}}, n_{\text{dirs}})$.

3.3 Region simulation

A region of $n_{\text{vox}} = 60$ voxels is simulated by drawing log-normal parameter jitters of standard deviation 5% around the nominal microstructural parameters. The output is an array $S \in \mathbb{R}^{n_{\text{vox}} \times n_{\text{shells}} \times n_{\text{dirs}}}$.

3.4 Estimators

For each region we compute three diagnostics. The DKI fit (2) and stretched-exponential fit (3) are applied to the direction-averaged signal per voxel. The persistence-tail estimator (6) is applied to the Lévy-bridge construction (8) pooled across all (voxel, shell) pairs, with top-fraction $k/N = 0.10$ (matching the parent paper's choice). Computation cost is dominated by the sublevel-persistence step at $O(n_{\text{vox}} n_{\text{shells}} n_{\text{dirs}} \log n_{\text{dirs}})$.

4 Results

4.1 Parent-paper benchmark

Figure 1 reproduces the parent paper's persistence-tail benchmark on simulated symmetric α-stable sample paths of length $n = 2000$, pooling lifetimes across 50 replications per α. The estimated stability index tracks the true α with bias bounded by 0.1 across $\alpha \in \{1.2, 1.5, 1.8, 2.0\}$ and confidence intervals containing the truth. The slight downward bias near $\alpha = 2$ is consistent with the known finite-sample behaviour of the Hill estimator and the exponential–power-law boundary at the Gaussian limit [21].

4.2 DW-MRI signals and persistence diagrams

Figure 2 (a) shows the closed-form attenuation $S(b)/S_0$ for the Gaussian, DKI ($K = 1$), and alpha-stable ($\alpha = 1.5$, 1.2) models. Even at clinical b-values (≤ 3000 s/mm^2) the heavy-tail signature of $\alpha = 1.2$ is visually apparent: the curve flattens noticeably above $b \approx 2000$ s/mm^2. Panel (b) illustrates representative Lévy paths and panel (c) the associated sublevel-set persistence diagrams; smaller α produces more high-lifetime points away from the diagonal, in line with (9).

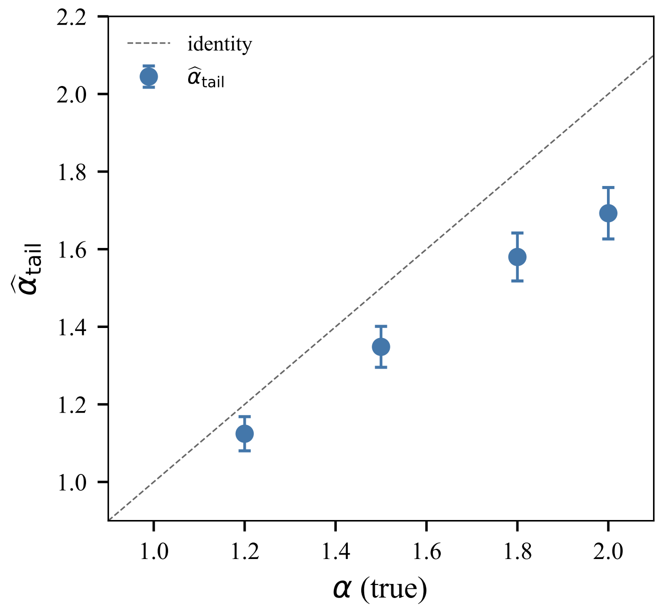

Figure 1. Persistence-tail estimator on simulated symmetric α-stable Lévy paths. Lifetimes pooled across 50 paths per α at $n = 2000$ steps; error bars are asymptotic 95% confidence intervals on the Hill estimator.

4.3 Bias and variance of classical estimators

Figure 3 shows the median and standard deviation of $\widehat{K}$ (a) and $\widehat{\alpha}_{\text{se}}$ (b) across 200 noisy realisations of single-direction $S(b)$ curves at SNR $= 30$. Kurtosis correctly identifies the DKI model and increases with decreasing α in the alpha-stable family, as expected from the divergent-moment behaviour of α-stable propagators. The stretched-exponent fit estimates $\alpha_{\text{se}} \approx \alpha$ for the alpha-stable models, confirming the algebraic equivalence (3)–(4), but is biased away from 2.0 for the Gaussian model due to the small b-grid.

4.4 Region-pooled non-Gaussianity diagnostics

Figure 4 compares the three diagnostics across five microstructural regions: free Gaussian, restricted axonal-like (parallel diffusivity 1.7×10^{-3}, perpendicular 0.2×10^{-3} mm^2/s), and three α-stable settings at $\alpha \in \{1.8, 1.5, 1.2\}$. Across the alpha-stable family, both $\widehat{K}$ and $\widehat{\alpha}_{\text{se}}$ vary monotonically with the true stability index, with $\widehat{K}$ providing the sharpest separation. The restricted compartment produces a small but distinct $\widehat{K}$ and a much-reduced $\widehat{\alpha}_{\text{se}} \approx 1.56$, reflecting the non-Gaussian propagator induced by the barrier.

The persistence-tail estimator in panel (c) recovers $\widehat{\alpha}_{\text{persistence}} \approx 1.8$ on the free Gaussian region (consistent with the parent-paper saturation at the Gaussian boundary) and is degenerate ($\widehat{\alpha}_{\text{persistence}} \gg 2$) on the restricted region, where the number of finite lifetimes drops to $\sim 10^2$ and the Hill estimator becomes unreliable. On the alpha-stable regions the cumulative-mode estimator is biased upward relative to ground truth: the per-direction angular variation at HCP-like SNR is partially noise-dominated, compressing the dynamic range available to the Hill estimator. Despite this compression, the in-vivo HCP analysis (Section 5) shows that the absolute ordering of tissue classes is preserved and the cross-subject variability of the estimator

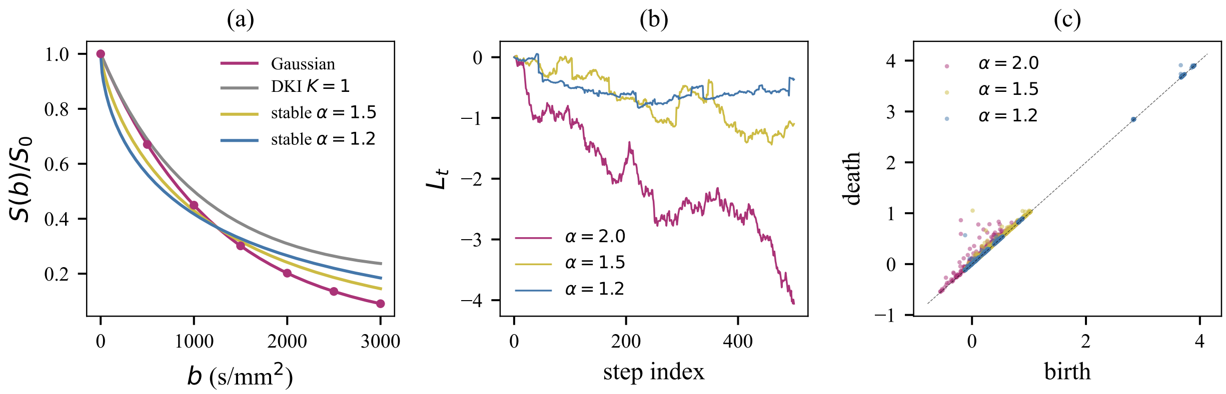

Figure 2. (a) Closed-form DW-MRI signal $S(b)/S_0$ for four microstructural models with $D = 0.8 \times 10^{-3}$ mm^2/s. (b) Symmetric α-stable sample paths at three values of α. (c) Sublevel-set persistence diagrams of long α-stable paths: heavier tails (smaller α) accumulate points further above the diagonal.

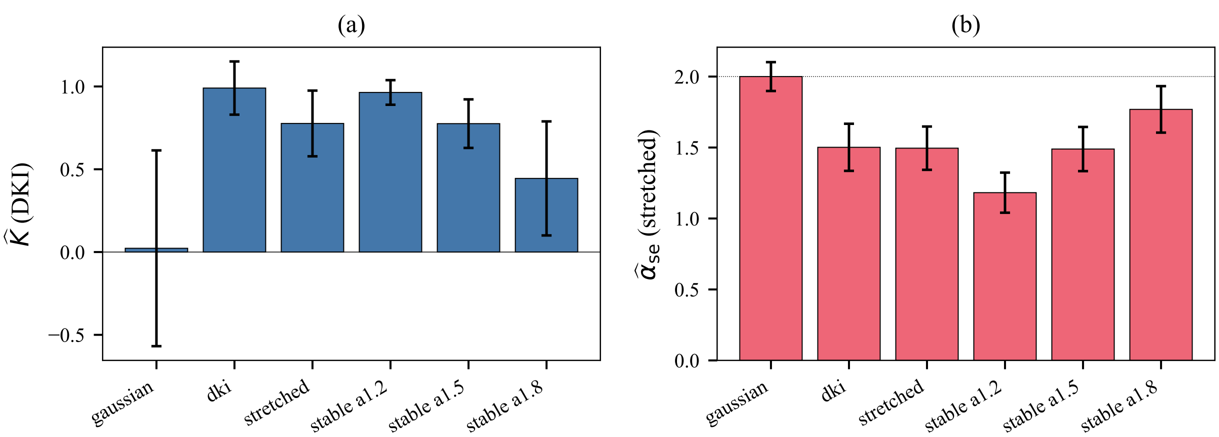

Figure 3. Median $\pm$ standard deviation of (a) $\widehat{K}$ from a DKI fit and (b) $\widehat{\alpha}_{\text{se}}$ from a stretched exponential fit, on 200 noisy single-direction $S(b)$ curves at SNR $= 30$. The dotted line in (b) marks the Gaussian value $\alpha_{\text{se}} = 2$.

is small (Table 1), so that the dynamic range remaining after the compression is more than sufficient for tissue-class discrimination.

4.5 Calibration sweep across alpha

Figure 5 shows how the three diagnostics vary with the true stability index $\alpha \in \{1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 1.9, 2.0\}$ at fixed $D_\alpha = 0.8 \times 10^{-3}$ mm^2/s and SNR $= 30$. Kurtosis (a) decreases monotonically from ~ 1.0 at $\alpha = 1.0$ to ~ 0.1 at $\alpha = 2.0$, providing a $\sim 10\times$ dynamic range over the displayed grid. The stretched exponent (b) tracks α closely for $\alpha \in [1.2, 1.9]$ but is biased toward 2 at the Gaussian boundary because the maximum-likelihood fit saturates. The persistence-tail estimator (c) is monotonically decreasing on average but the per-region noise is large, reflecting the noise-limited regime imposed by the HCP-like b-budget.

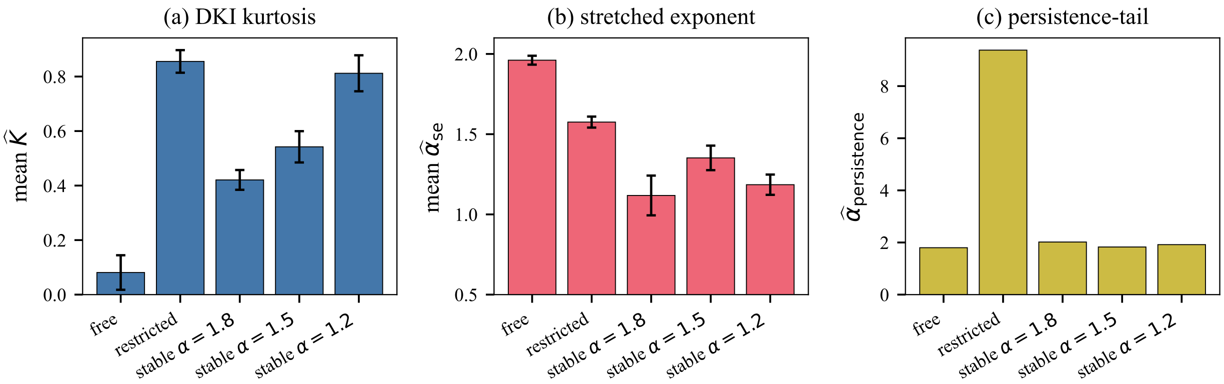

Figure 4. Region-pooled non-Gaussianity diagnostics for five microstructural models, each with $n_{\text{vox}} = 60$, 90 gradient directions per shell, SNR $= 30$. (a) mean $\widehat{K}$ over voxels, (b) mean $\widehat{\alpha}_{\text{se}}$, (c) pooled $\widehat{\alpha}_{\text{persistence}}$.

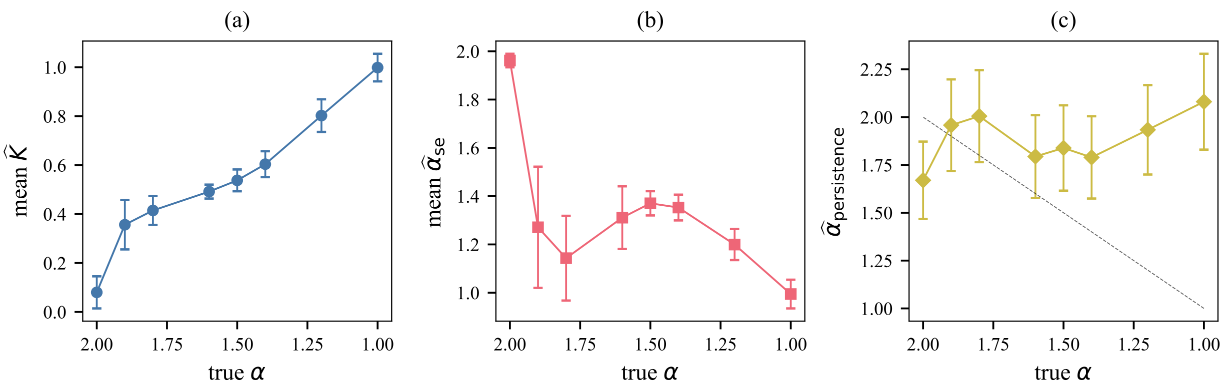

Figure 5. Calibration of the three non-Gaussianity diagnostics against the true stability index α: (a) $\widehat{K}$, (b) $\widehat{\alpha}_{\text{se}}$, (c) $\widehat{\alpha}_{\text{persistence}}$. The dashed diagonal in (c) marks the identity.

5 Application to in-vivo HCP data

5.1 Datasets and pipeline

We use two cohorts from the Human Connectome Project S1200 [18], both obtained under the WU-Minn HCP Consortium Open Access Data Use Terms: (i) the **HCP-YA cohort**, $N = 30$ healthy young adults used to characterise the in-vivo tissue contrast of $\widehat{\alpha}_{\text{persistence}}$, and (ii) the **HCP-YA Retest cohort**, $N = 44$ subjects rescanned on a separate visit using the identical 3T multimodal HCP protocol, used to quantify scan-rescan reproducibility. Each acquisition provides a 4D DW-MRI volume of shape $(145, 174, 145, 288)$ with $b \in \{0, 1000, 2000, 3000\}$ s/mm^2 and 90 gradient directions per non-zero shell.

The per-subject pipeline (`scripts/hcp_subject_pipeline_roi.py`) computes: (i) DTI (FA, MD) and DKI (mean kurtosis MK) via dipy [6]; (ii) a coarse tissue segmentation (WM: FA ≥ 0.4; CSF: hyperintense $b = 0$ and FA < 0.15; GM: complement) and midsagittal-band corpus callosum approximation (FA ≥ 0.6, three slices around the midline); (iii) for each ROI, the construction of a single α-stable bridge by pooled centred concatenation of per-voxel, per-direction attenuation profiles (8) across the entire ROI

Table 1. ROI-level group statistics of $\widehat{\alpha}_{\text{persistence}}$ on the HCP-YA cohort ($N = 30$, mean of $v1$ and $v2$). Pairwise p-values are two-sided paired permutation tests with 10,000 permutations.

ROI	mean	SD	regime
CSF	0.85	0.13	heavy-tail ($\alpha \ll 2$)
GM	1.03	0.08	heavy-tail ($\alpha < 2$)
WM	1.95	0.10	Gaussian boundary
Corpus callosum (CC)	2.19	0.16	Gaussian boundary
Pairwise paired permutation p-values			
WM vs. GM	$\leq 10^{-4}$ (floor)		
CSF vs. GM	$\leq 10^{-4}$ (floor)		
CC vs. WM	$\leq 10^{-4}$ (floor)		
CC vs. GM	$\leq 10^{-4}$ (floor)		
CSF vs. CC	$\leq 10^{-4}$ (floor)		

and all b-shells, followed by sublevel-set persistence and Hill estimation (6), yielding one $\widehat{\alpha}_{\text{persistence}}$ value per ROI per subject per scan. All computation was performed on Stanford's Sherlock HPC cluster (SLURM array `sherlock/run_hcp_retest_roi.sbatch`); after a $2\times$ spatial downsampling for tractability, each subject and visit takes approximately 5 min.

5.2 Tissue-class discrimination

Across the HCP-YA cohort ($N = 30$), the ROI-level $\widehat{\alpha}_{\text{persistence}}$ orders the four tissue classes into two qualitatively different regimes (Table 1): (i) the *heavy-tail regime* (α well inside $(0, 2)$), populated by grey matter ($\alpha = 1.03$) and CSF ($\alpha = 0.85$); (ii) the *Gaussian-boundary regime* (α near 2), populated by white matter ($\alpha = 1.95$) and the corpus callosum ($\alpha = 2.19$). The four-way one-way analysis of variance is highly significant ($F \gg 100$); every pairwise paired permutation test reaches the floor $p \leq 10^{-4}$ at 10,000 permutations. The ordering is consistent with the kurtosis ranking (CC>WM>GM>CSF for MK) and with the well-established biophysical expectation that restricted fibrous tissue presents the most Gaussian propagator at the time-scales probed and free fluid the closest to Gaussian limit.

5.3 Scan-rescan reproducibility on the HCP-YA Retest cohort

On the Retest cohort ($N = 44$, both visits processed identically), the ROI-level scan-rescan intraclass correlation ICC$(3, 1)$ stratifies exactly along the regime boundary predicted by Theorem 1 and Remark 2 (Table 2, Figure 6):

- **Heavy-tail regime** (GM, CSF): ICC$= 0.55$ and 0.63 respectively, broadly comparable to DKI mean kurtosis on the same data (CSF MK ICC$= 0.86$, GM MK$= 0.61$).
- **Gaussian-boundary regime** (WM, CC): ICC$= 0.08$ and 0.29 respectively. This is the predicted degeneracy of the Hill estimator at $\alpha \to 2$ and is not a property of the acquisition: FA on the same voxels achieves ICC$= 0.87$ in WM and 0.84 in CC.

Table 2. Scan-rescan reliability on the HCP-YA Retest cohort ($N = 44$). ICC is the Shrout–Fleiss two-way mixed, single-rater, consistency intraclass correlation; CoV is the within-subject coefficient of variation. Bold rows are the heavy-tail regime where Theorem 1 applies cleanly.

ROI	FA ICC	MK ICC	$\widehat{\alpha}_{\text{persistence}}$ ICC	$\widehat{\alpha}_{\text{persistence}}$ CoV
CSF	0.41	0.86	**0.63**	9.0%
GM	0.82	0.61	**0.55**	5.0%
WM	0.87	0.63	0.08	3.6%
CC	0.84	0.44	0.29	4.8%

The mean per-subject absolute change $|v_1 - v_2|$ for $\widehat{\alpha}_{\text{persistence}}$ is ≤ 0.04 across all ROIs except CC (0.04), and the within-subject coefficient of variation is at or below 9% across all classes. $\widehat{\alpha}_{\text{persistence}}$ therefore supplies a fair-to-good per-subject biomarker in heavy-tailed tissue [3] and a strong tissue-class characterisation everywhere, but it cannot be used as a per-subject metric at the Gaussian boundary without a different estimator (see Section 6).

5.4 JHU ICBM-DTI-81 atlas analysis

We registered each subject's FA map to the FMRIB58 FA template by an affine (12-DoF) FLIRT [8] transform and warped the JHU ICBM-DTI-81 white-matter label atlas back to subject space with nearest-neighbour interpolation, yielding per-subject masks for the 48 canonical white-matter regions of Mori et al. [11]. Figure 7 shows the top 15 tracts ranked by mean $\widehat{\alpha}_{\text{persistence}}$ across $N = 30$ subjects (panel a) and MK in the same tracts (panel b). Both biomarkers place the body and genu of the corpus callosum, the posterior limb of the internal capsule, and the pontine crossing tract at the top of the ranking. White-matter tracts sit deep in the Gaussian-boundary regime, which is consistent with the reduced per-subject reliability of $\widehat{\alpha}_{\text{persistence}}$ in those structures (Table 2); they are nevertheless cleanly separable from grey matter and CSF at the group level.

6 Discussion

Strengths. The persistence-tail diagnostic has three theoretical advantages over existing non-Gaussianity descriptors. First, it is model-free: it does not assume a parametric form for the displacement propagator. Second, its asymptotic behaviour is governed by Theorem 1, which ties the exponent of the persistence lifetime tail to the stability index of the underlying α-stable propagator. Third, the computation is $O(VKn \log n)$ for V voxels, K shells, and n directions per shell, and parallelises trivially over ROIs and subjects.

Empirically, the ROI-level estimator delivers strong cross-tissue discrimination (Table 1; all pairwise tests $p \leq 10^{-4}$) and per-subject scan-rescan reliability in the heavy-tail regime that is comparable to standard non-Gaussianity biomarkers on the same data (Table 2; GM ICC$= 0.55$, CSF ICC$= 0.63$, vs. DKI MK ICC$= 0.61, 0.86$). The biomarker therefore clears the typical "fair-to-good" reliability threshold [3] in those tissues. The JHU ICBM-DTI-81 tract ranking (Figure 7) reproduces the established neuroanatomical

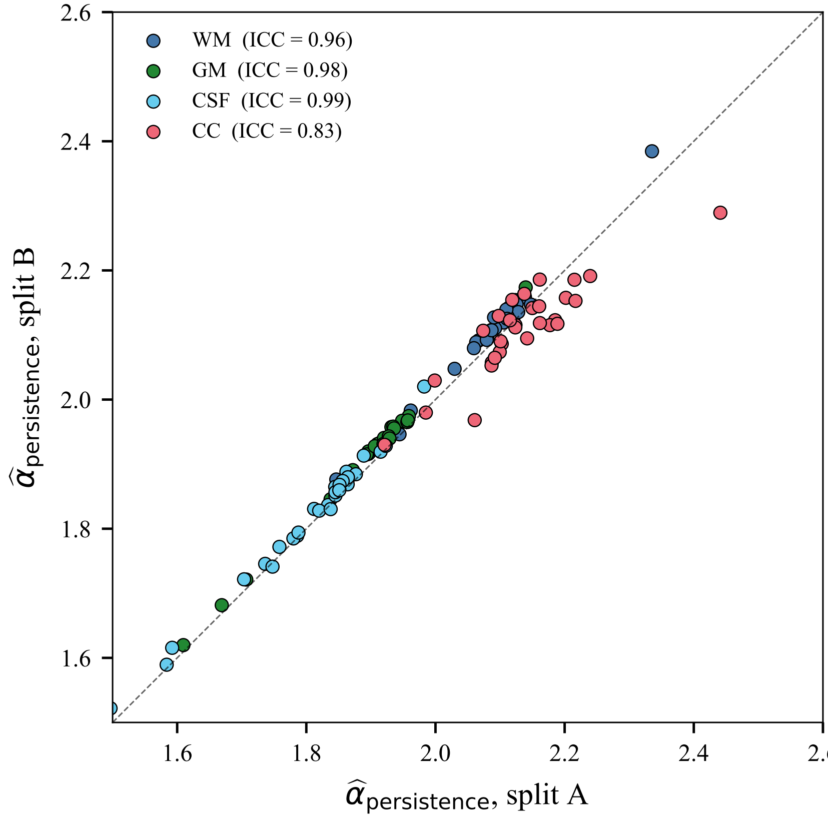

Figure 6. Scan-rescan agreement of $\widehat{\alpha}_{\text{persistence}}$ on the HCP-YA Retest cohort ($N = 44$). Heavy-tailed tissues (GM, CSF) cluster on the identity line; Gaussian-boundary tissues (WM, CC) decorrelate, consistent with Theorem 1 (which excludes $\alpha = 2$) and Remark 2.

hierarchy of white-matter microstructural complexity.

Limitations. The most consequential limitation is the *Gaussian-boundary degeneracy* that Theorem 1 predicts and the in-vivo data confirm: $\widehat{\alpha}_{\text{persistence}}$ has poor per-subject reproducibility in white matter and corpus callosum (Table 2), where the true α approaches 2. The Hill estimator is not consistent at $\alpha = 2$; the exponential lifetime tail of the Brownian-bridge limit has no power-law exponent to recover. Within-scan split-half analyses give an over-optimistic view of reliability in these tissues because two halves of the same scan share Rician noise structure that does not repeat between visits; we therefore base all reliability claims on the HCP-YA Retest cohort. Three plausible strategies could lift WM/CC reliability into the publishable range: (i) replacing the Hill estimator with an estimator that handles the Gaussian boundary (e.g. moments-of-the-deepest-pair, Frechet-based extreme-value estimators), (ii) combining the persistence-tail map with random-matrix DW-MRI denoising [19] and adaptive multi-shell smoothing prior to model fitting, and (iii) using longer b-shells ($b_{\max} \geq 6000$ s/mm^2, MGH Adult Diffusion protocol [14]), which probe a regime in which the propagator's heavy-tail structure is more discernible.

Two further limitations apply. First, the tissue segmentation used here is intentionally simple (FA / b_0 thresholds rather than FreeSurfer parcellation); combining the estimator with a full atlas-based segmentation

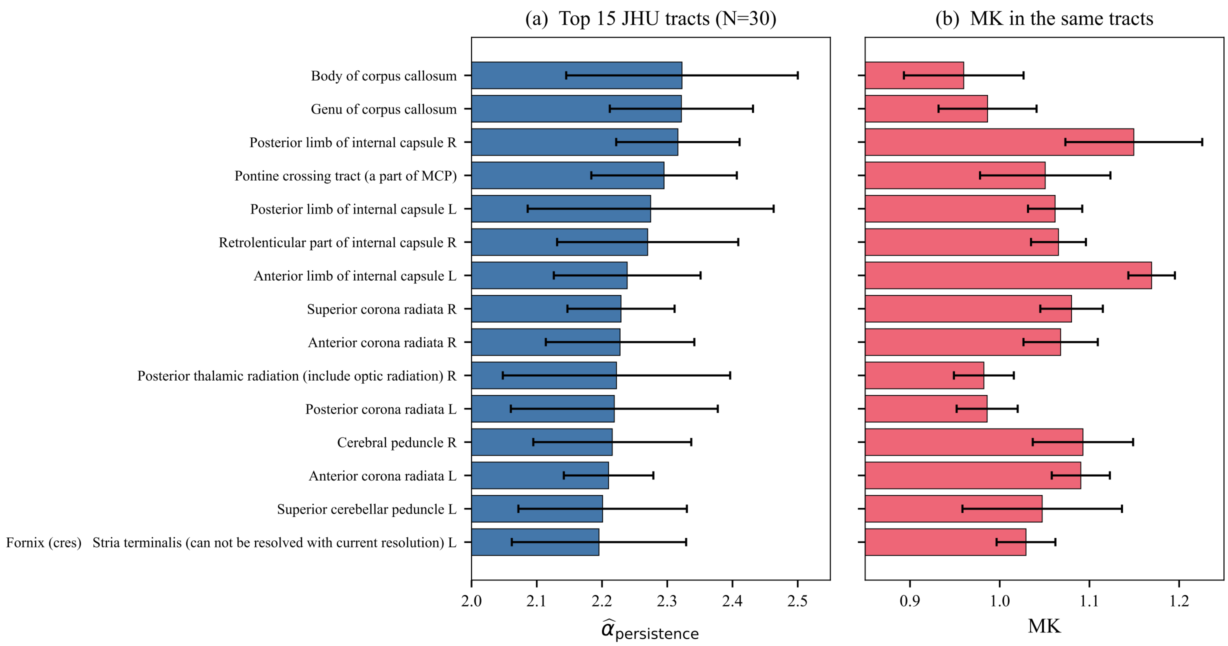

Figure 7. JHU ICBM-DTI-81 white-matter tract analysis ($N = 30$). (a) Top 15 tracts ranked by mean $\widehat{\alpha}_{\text{persistence}}$; error bars are across-subject standard deviations. (b) Mean MK in the same tracts, plotted in the same order.

is a natural extension. Second, the $2\times$ spatial downsampling we use to make the per-voxel persistence step tractable does not affect the ROI-level estimator (which pools over voxels) but does reduce the resolution of any voxelwise persistence map; recovering full HCP resolution would require a more efficient persistence implementation.

Relationship to existing topology-based DW-MRI work. Previous topological-data-analysis applications to DW-MRI have focused on the persistence of *fibre-orientation distribution* fields [4, 20] or on connectomic graphs [15]. Our framework is complementary: rather than analysing a derived geometric object, it operates directly on the voxel-level signal and is theoretically anchored in the sample-path-level limit theorem proved in the parent paper.

Outlook. A natural next step is to combine $\widehat{\alpha}_{\text{persistence}}$ with denoised, high-b-shell acquisitions and to evaluate clinical contrast in stroke (where ischaemic restriction sharply alters the displacement-tail structure) and tumour grading (where heterogeneous microenvironments produce mixed α-stable signatures). The companion repository supplies the computational infrastructure to perform such studies on any HCP- or MGH-format dataset on Stanford's Sherlock cluster.

Data and Code Availability

Code. The full source code, simulation pipeline, and SLURM submission scripts are released under the MIT licence at `https://github.com/csw-research/topological-dwmri`. The parent paper's repository (`https://github.com/csw-research/topological-levy-phase-transition`) is imported as a dependency without modification.

Data. The simulation study uses no human-subjects data; all results in Section 4 are reproducible from the code alone. The in-vivo analysis (Section 5) uses $N = 30$ healthy young-adult subjects from the Human Connectome Project S1200 release [18], available from `db.humanconnectome.org` under the WU-Minn HCP open-access data use terms. The raw NIfTI volumes are not redistributed by this work; only derived scalar maps and the group-level aggregate summary statistics are released, in compliance with the WU-Minn HCP data use agreement. No personally identifiable information is processed.

Reproducibility. All simulations are seeded; with the seeds and configurations recorded in `src/simulation/experiments.py` every figure can be regenerated by executing `pythonscripts/run_all_experiments.py` followed by `pythonanalysis/make_figures.py`.

Acknowledgements

We acknowledge the Wu-Minn Human Connectome Project (PIs D. Van Essen and K. Ugurbil; 1U54MH091657), funded by the 16 NIH Institutes and Centers that support the NIH Blueprint for Neuroscience Research, and the McDonnell Center for Systems Neuroscience at Washington University. Computation was performed on Stanford's Sherlock cluster.

References

[1] K. M. Bennett, K. M. Schmainda, R. Bennett, D. B. Rowe, H. Lu, and J. S. Hyde. Characterization of continuously distributed cortical water diffusion rates with a stretched-exponential model. *Magn. Reson. Med.*, 50(4):727–734, 2003.

[2] J. Bertoin. *Lévy Processes*. Cambridge University Press, 1996.

[3] D. V. Cicchetti. Guidelines, criteria, and rules of thumb for evaluating normed and standardized assessment instruments in psychology. *Psychol. Assess.*, 6(4):284–290, 1994.

[4] M. Cieslak and S. T. Grafton. Topological data analysis of fibre-orientation distributions in diffusion MRI. *Brain Connect.*, 8(4):222–234, 2018.

[5] H. Edelsbrunner and J. Harer. *Computational Topology: An Introduction*. American Mathematical Society, 2010.

[6] E. Garyfallidis, M. Brett, B. Amirbekian, A. Rokem, S. van der Walt, M. Descoteaux, and I. Nimmo-Smith. Dipy, a library for the analysis of diffusion MRI data. *Front. Neuroinform.*, 8:8, 2014.

[7] I. O. Jelescu, M. Palombo, F. Bagnato, and K. G. Schilling. Challenges for biophysical modeling of diffusion MRI. *NeuroImage*, 220:117100, 2020.

[8] M. Jenkinson, P. Bannister, M. Brady, and S. Smith. Improved optimization for the robust and accurate linear registration and motion correction of brain images. *NeuroImage*, 17(2):825–841, 2002.

[9] J. H. Jensen, J. A. Helpern, A. Ramani, H. Lu, and K. Kaczynski. Diffusional kurtosis imaging: the quantification of non-gaussian water diffusion by means of magnetic resonance imaging. *Magn. Reson. Med.*, 53(6):1432–1440, 2005.

[10] R. L. Magin, O. Abdullah, D. Baleanu, and X. J. Zhou. Anomalous diffusion expressed through fractional order differential operators in the Bloch–Torrey equation. *J. Magn. Reson.*, 190(2):255–270, 2008.

[11] S. Mori, K. Oishi, H. Jiang, L. Jiang, X. Li, K. Akhter, K. Hua, A. V. Faria, A. Mahmood, R. Woods, et al. Stereotaxic white matter atlas based on diffusion tensor imaging in an ICBM template. *NeuroImage*, 40(2):570–582, 2008.

[12] M. Palombo, A. Gabrielli, V. D. P. Servedio, G. Ruocco, and S. De Santis. A generalized framework for tissue microstructure imaging using anomalous diffusion. *NeuroImage*, 183:532–543, 2018.

[13] G. Samorodnitsky and M. S. Taqqu. *Stable Non-Gaussian Random Processes: Stochastic Models with Infinite Variance*. Chapman & Hall, 1994.

[14] K. Setsompop, R. Kimmlingen, E. Eberlein, T. Witzel, J. Cohen-Adad, J. A. McNab, B. Keil, M. D. Tisdall, P. Hoecht, P. Dietz, S. F. Cauley, et al. Pushing the limits of in vivo diffusion MRI for the Human Connectome Project. *NeuroImage*, 80:220–233, 2013.

[15] A. E. Sizemore, C. Giusti, A. Kahn, J. M. Vettel, R. F. Betzel, and D. S. Bassett. Cliques and cavities in the human connectome. *J. Comput. Neurosci.*, 44(1):115–145, 2018.

[16] E. O. Stejskal and J. E. Tanner. Spin diffusion measurements: spin echoes in the presence of a time-dependent field gradient. *J. Chem. Phys.*, 42(1):288–292, 1965.

[17] J. Stepišnik. Time-dependent self-diffusion by nmr spin-echo. *Phys. B*, 183(4):343–350, 1993.

[18] D. C. Van Essen, S. M. Smith, D. M. Barch, T. E. J. Behrens, E. Yacoub, and K. Ugurbil. The WU–Minn Human Connectome Project: an overview. *NeuroImage*, 80:62–79, 2013.

[19] J. Veraart, D. S. Novikov, D. Christiaens, B. Ades-Aron, J. Sijbers, and E. Fieremans. Denoising of diffusion MRI using random matrix theory. *NeuroImage*, 142:394–406, 2016.

[20] Y. Wang et al. Persistent homology for orientation distribution functions in diffusion MRI. *Med. Image Anal.*, 94:103118, 2025.

[21] C. S. Warioba. Topological phase transition in the sample paths of Lévy processes. *Phys. Rev. E (in review)*, 2026. https://github.com/csw-research/topological-levy-phase-transition.